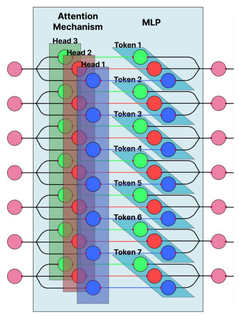
LIG: Layer-wise Integrated Gradients for Within-Layer Flow Analysis in Transformers

MON-19

Poster (PDF) · My page

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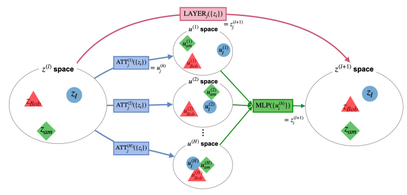
Problem



Attention maps miss what values and the MLP pass on.

They show where a layer looks, not what it passes on.

Idea OURS



$$\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j} \overset{?}{\approx} \sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}$$

whole layer | composed

Measure ATT and MLP separately with IG; test their composition.

Set-to-set Integrated Gradients (IG) at the Multi-Head Attention (ATT) and MLP boundaries, composed and compared with whole-layer IG.

Result

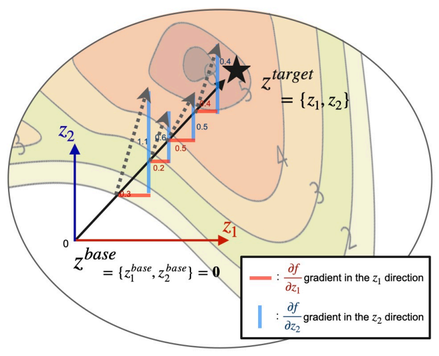
Baselines matched to the whole-layer one gave the smallest composition gap.

BERT-base, 1,700 Penn Treebank (PTB) sentences; within each reference (ITB, Zero), among tested configurations.

Agreement between attributions, not behavioural faithfulness; the composition omits the residual route (§5).

Project page
https://eightsuzuki.github.io/layer-wise-integrated-gradients/

1 IG, and where the path starts



PRIOR IG [1] *line-integrates* the gradient along a path from a **baseline** $z(0)$ to the input z. Any path keeps completeness, so path and start are ours to choose; we use the straight path $z(a) = z(0) + a\,(z - z(0))$:

$$\mathrm{IG}_d = (z_d - z_d(0)) \int_0^1 \frac{\partial f(z(a))}{\partial z_d}\, da$$

The baseline sets what each contribution is measured against.

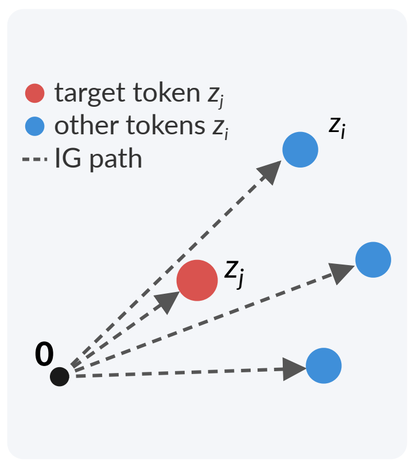
2 LIG: set-to-set IG at each boundary OURS

At each boundary: IG of output token j's L_2 change, per input token or head (§6):

ATT	MLP (W_o, LayerNorm, FFN; residual z fixed)	Whole layer l
$z_i \to u_j^{(h)}$: $\mathrm{IG}^{\mathrm{ATT}}_{i,j,h}$	$u_j^{(h)} \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{MLP}}_{h,j}$	$z_i \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{LAYER}}_{i,j}$

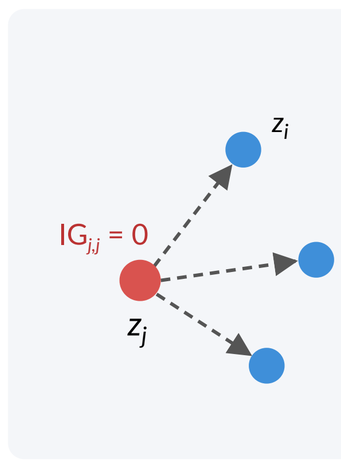
Baselines: how the **input** moves along the IG path, $a = 0 \to 1$.

Zero



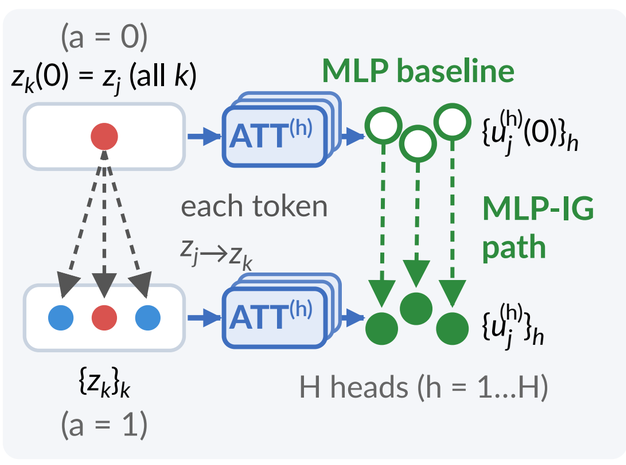
Every path starts at the zero vector $\mathbf{0}$.

Input Token Base (ITB)



All tokens start at the target z_j; z_j itself does not move, so $\mathrm{IG}_{j,j} = 0$.

MLP: ATTITBa=0

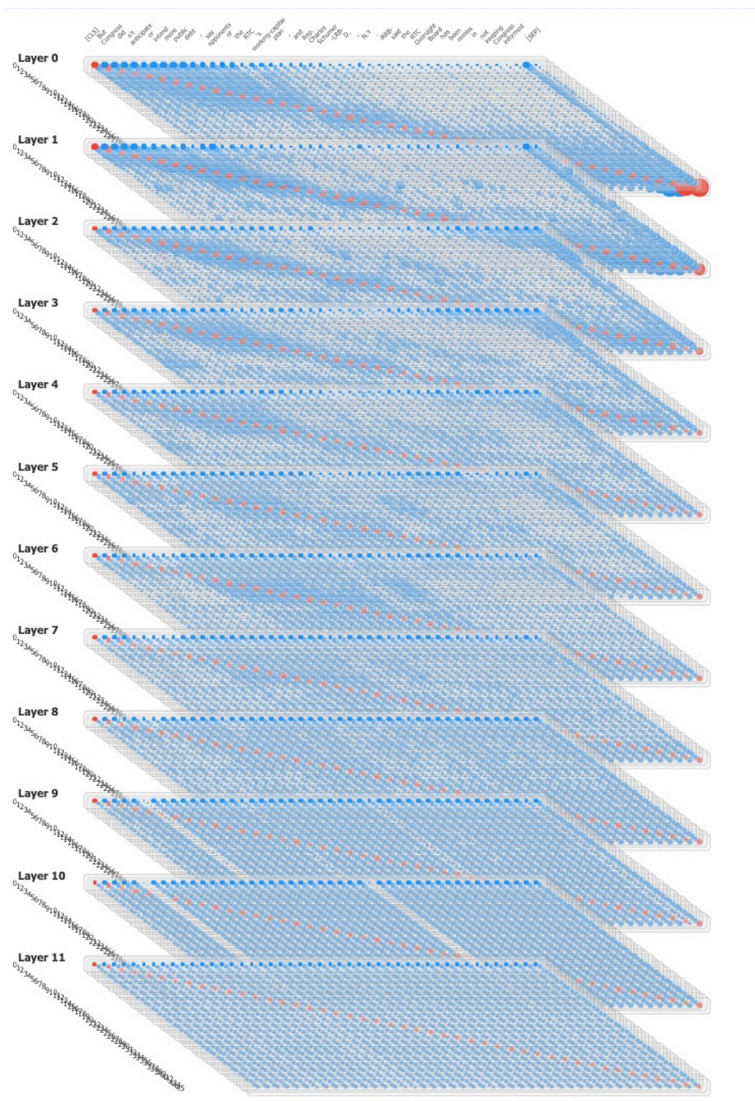


At $a = 0$ all tokens are z_j, so each head outputs z_j's value $u_j^{(h)}(0)$: the MLP baseline.

Why ITB? ATT weight-averages token values, so we measure each token *relative to the target* z_j; its zero self term [8] comes from the reference, not the layer. **Why ATTITBa=0?** MLP inherits the ATT output at the ITB start ($a = 0$), keeping both modules on one reference.

3 Use case: the flow layer by layer

Whole-layer IG, Zero baseline, one PTB sentence (qualitative). Columns: input tokens; rows: output tokens; circle size = |contribution|; red = self, blue = other tokens. **Layer 0:** local; **layers 9–11:** sentence-wide.



4 Result: does the composition match the layer?

Setup. BERT-base-uncased; PTB dev, 1,700 sentences [9]; $m = 32$ steps. Per sentence, layer l and output token j: the L_2 distance between the two sides of the Idea formula, as vectors over i (lower = better). Top 3 per block; **bold** = best; **compare within blocks**.

Whole-layer baseline (reference)	Composed: ATT	MLP	L_2 ↓
ITB Input Token Base: all tokens start at z_j	**ITB**	**ATTITBa=0**	**0.305**
	ITB-zeroRatio	ATTITBa=0	0.312
	ITB-mapRatio	ATTITBa=0	0.318
Zero zero vector $\mathbf{0}$	**Zero**	**Zero**	**0.736**
	ITB-zeroRatio	Zero	0.744
	Zero	ATTITBa=0	0.749

> **The best MLP baseline follows the reference: ATTITBa=0 under ITB (for every tested ATT baseline); Zero under Zero.**

5 Composing ATT and MLP OURS

The composition (top, Idea) follows the spirit of LRP [2, 3]. **Not an IG chain rule:** it omits the residual route, and separately integrated paths need not agree. We measure this *compositional gap*.

6 Details and prior work

Scalarize by $f = \|y_j(a) - y_j(0)\|_2$ and sum IG within each input element r; scores are signed and normalized:

$$\mathrm{IG}_{r,j} = \sum_{d \in r} \mathrm{IG}_d, \qquad \widetilde{\mathrm{IG}}_{r,j} = \frac{\mathrm{IG}_{r,j}}{\|y_j(1) - y_j(0)\|_2}$$

ITB's zero self term is completed by zeroRatio (from Zero-baseline scores) or mapRatio (from $\alpha_{j,j}$), keeping the total.

Method	Limitation / LIG's gain
PRIOR Attention map [7]	ignores values and MLP
PRIOR Rollout [5]	ignores values and MLP
PRIOR Norm-based [6]	no ATT / MLP split
PRIOR AttnLRP [4]	a rule per operation
OURS **LIG**	**one rule (IG); splits and composes**

7 Limits & next steps

- **Limits:** BERT-base / PTB only; straight paths may leave the data distribution; ITB completion uniqueness is open; L_2 scalarization discards direction.
- **Next:** decoder-only models (LIG needs only module boundaries and gradients); EAP / EAP-IG [10, 11]; intervention-based checks.

[1] Sundararajan+ ICML 2017 · [2] Bach+ PLOS ONE 2015 · [3] Montavon+ LRP overview 2019 · [4] Achtibat+ (AttnLRP) ICML 2024 · [5] Abnar & Zuidema ACL 2020 · [6] Kobayashi+ EMNLP 2020 · [7] Jain & Wallace NAACL 2019 · [8] Sturmfels+ Distill 2020 · [9] Marcus+ Treebank-3, LDC 1999 · [10] Syed+ BlackboxNLP 2024 · [11] Hanna+ COLM 2024. Supported by JSPS KAKENHI JP26K02989, JP26K23861, JP25K24611.